Introduction

In this project we use Bayesian hierarchical models to analyze self-reported happiness, measured by the Cantril ladder, at the country, continent and global levels. We analyze global happiness trends, evaluate global happiness in 2025, rank countries by happiness and compute the probability that Slovenia is top 20 rank, and lastly we investigate whether wealth or health have a larger positive impact on happiness, per continent.

Methods

Data

The dataset consists of N observations with a continuous target variable $y_i \in [0,10]$, indicating an individual's self-reported happiness score on the Cantril scale. Each observation also has several dependent variables, describing the respondent's environment, including country, continent, log-transformed GDP per capita, life expectancy and the year the answer was given.

The dataset consists of 6 different continents, 165 countries and 1956 entries. The mean year is 2017 and mean life expectancy is 73. If we assume the GDP was log-transformed using the natural logarithm, the original mean GDP was around 25 000 USD.

Before fitting the model, the year was centered and life expectancy and log GDP were standardized.

Modeling

Let $i = 1,\dots,N$ be observation indices, $c = 1,\dots,C$ countries and $k = 1,\dots,K$ continents. We modeled the happiness score using linear regression:

$$y_i \sim N(\mu_i, \sigma_y)$$

where

$$\mu_i = \alpha_{c[i]} + \sum_{\text{feat}\in\{\text{year, logGDP, lifeExp}\}} \beta_{c[i],\text{feat}} \cdot \text{feat}_i$$

Country-level parameters are modeled hierarchically as draws from continent-level distributions

$$\alpha_c \sim N(\alpha_{k[c]}, \sigma_{\alpha,\text{country}})$$

$$\beta_c \sim N(\beta_{k[c]}, \sigma_{\beta,\text{country}})$$

where β_c is the vector of regression coefficients for country c.

Similarly, continent-level parameters are modeled using global priors

$$\alpha_k \sim N(\alpha_{\text{global}}, \sigma_{\alpha,\text{continent}})$$

$$\beta_k \sim N(\beta_{\text{global}}, \sigma_{\beta,\text{continent}})$$

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Global priors are set to $\alpha_{global} \sim N(5,2)$ and $\beta_{global} \sim N(0,1)$. Exponential(1) hyperpriors were placed on all variance parameters.

The model fit well, traceplots show no reason for concern, $\hat{r}$ and ESS for the α and β parameters on country, continent and global scope were 1 and over 400 for the majority of the parameters, however some less-represented countries had a lower ESS.

Results

Global happiness trend

First we investigated if happiness is globally positively correlated with the year. This was done by analyzing the posterior distribution of the global year coefficient, $\beta_{\text{year, global}}$. The posterior probability that this coefficient is positive is 0.64 ± 0.01, indicating moderate evidence that the trend is positive.

Are humans happy in 2025 from the global perspective?

Next we investigated whether global happiness in 2025 exceeds a neutral reference level. Using the neutral Cantril ladder score of 5 as a baseline, we evaluated posterior predictive mean happiness for 2025 by combining the global intercept with the global year effect, β_{year}. The posterior probability that the predicted global mean happiness is larger than 5 is 0.95 ± 0.01, indicating strong evidence, that the average global happiness in 2025 is above the neutral level.

What is the probability that Slovenia is ranked in the top 20 countries happiness wise in 2025?

We also analyzed the probability of Slovenia being ranked top 20 happiness wise in 2025. This was done by computing posterior predictive happiness scores for all countries in 2025, ranking them and estimating the posterior probability that Slovenia is in the top 20 ranks. We used average GDP and life expectancy values for each country. Specifically, we computed

$$\mu_c = \alpha_c + \beta_{c,\text{year}} * 2025_{\text{centered}} + \beta_{\text{c, logGDP}} * \overline{\text{logGDP}}_c + \beta_{\text{c, lifeExp}} * \overline{\text{lifeExp}}_c,$$

and then ranked each μ_c and computed the probability that $\mu_{\text{Slovenia}} \in$ Top 20. The resulting posterior probability is 0.36 ± 0.01. Figure 1 shows the posterior distribution of Slovenia's rank and the cutoff. Notice that the distribution looks like a skewed normal with a longer-than-usual tail with the mean at rank 23.

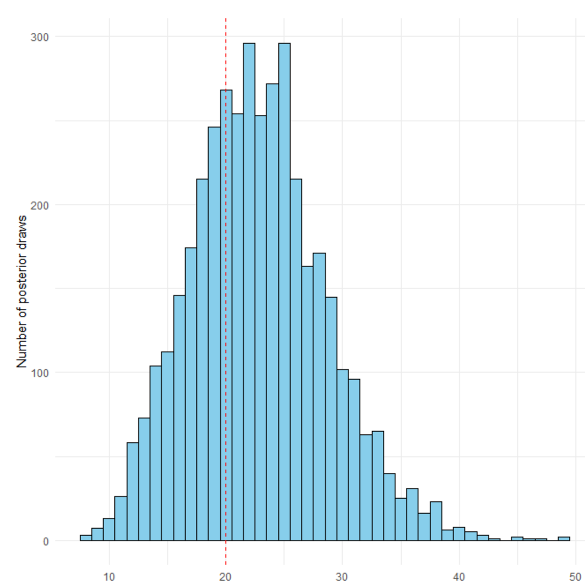

Figure 1. Posterior distribution of Slovenia's rank in 2025 among all countries. The dashed red line shows the cutoff at rank 20. The mode is at rank 22.

Health vs. wealth - effects on happiness

Finally we investigated whether there were any major differences between the correlation of health and wealth to happiness, on a continent scale. To compare the effects of health and wealth on happiness on continent level, we analyzed posterior draws of the corresponding coefficients.

Figure 2 shows the posterior continent-level coefficient draws. Notice that except for North America, the mean wealth effect is consistently more positively correlated to the happiness score than the mean health effect. Table 1 shows the life expectancy and GDP mean effects, along with the probability that GDP has a higher positive effect on happiness than health. Except for Oceania and North America, the positive correlation between wealth and happiness is larger than between health and happiness with a probability over 90%. Asia shows the largest difference between the two, having almost perfect separation of the distributions. The country with the closest impact of both is North America, which can probably be attributed to the fact that health in North America is closely tied to personal finances, since individuals often have to pay out-of-pocket for medical care or rely on private insurance coverage.

Figure 2 also illustrates the differences in the uncertainty across the continents. Asia, for example has relatively narrow distributions, showing that wealth and health both have approximately the same effect on all the respondees, while South America and Oceania show high uncertainty, indicating that while being wealthy means being way happier for some, but can have very little effect on others.

Continent	Mean LE effect	Mean GDP effect	P(LE < GDP)
Africa	0.150	**0.416**	0.91 ± 0.02
Asia	0.017	**0.819**	0.99 ± 0.00
Europe	0.359	**1.060**	0.93 ± 0.02
N. America	0.492	**0.598**	0.60 ± 0.04
Oceania	0.353	**0.833**	0.80 ± 0.01
S. America	0.058	**0.710**	0.91 ± 0.01

Table 1. Mean effects of life expectancy (LE) and GDP by continent, with probability that GDP has a larger effect than LE. Bold indicates the higher effect in each row.

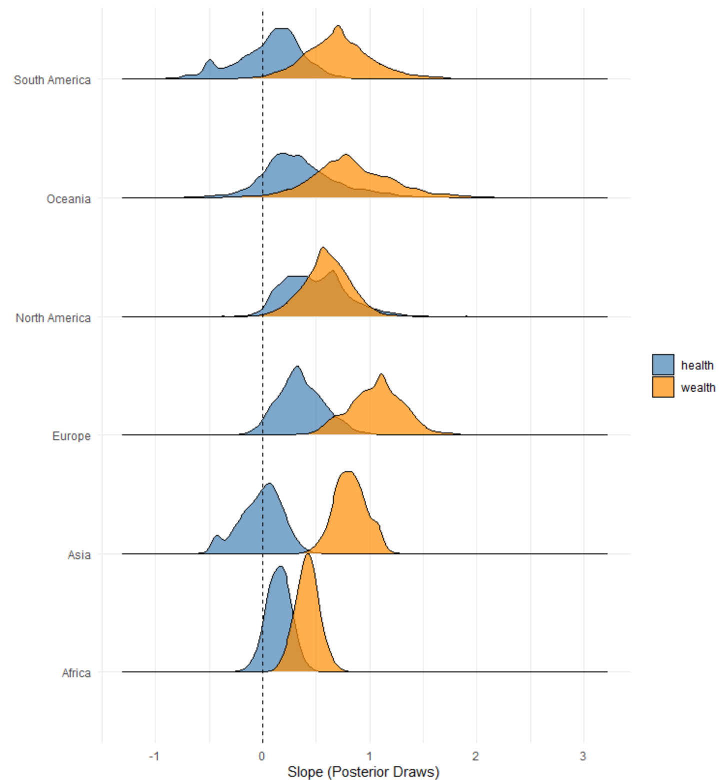

Figure 2. Posterior distributions of continent-level slopes: health vs. wealth. Notice that the mean effect of wealth is consistently more positively correlated to happiness than the mean effect of health.

Discussion

Using hierarchical Bayesian models allowed us to look at happiness at multiple levels—global, continent, and country—while accounting for differences between countries and continents. It also has the Bayesian shrinkage property, which is very important in our case, since some countries had very little data points.

Globally, happiness seems to be slightly increasing over the years, with a 64% posterior probability that the trend is positive. For 2025, the predicted global mean happiness is above the neutral level of 5, with high confidence. Looking at specific countries, Slovenia has only a moderate probability (36%) of being in the top 20 happiest countries in 2025, suggesting that while it is relatively happy, it is not among the very top.

At the continent level, wealth (GDP) generally has a stronger effect on happiness than health (life expectancy). The table and posterior distributions show that except for North America and Oceania, the probability that GDP has a larger effect than LE is over 90%. Asia shows the clearest separation, meaning that income seems much more important for happiness there than health. North America has the closest effects for GDP and LE, probably because health is tied to personal finances in the U.S. and Canada, so the two factors overlap. Some continents, like Oceania and South America, show wider posterior distributions, indicating more uncertainty and variation in how wealth and health affect happiness for different individuals.

Overall, the analysis shows that wealth tends to dominate over health in predicting happiness, but there are interesting regional differences.